UNIT - I

Introduction, Design for Static and Dynamic Loads

1. Mechanical Engineering Design

Design = creative decision-making process that converts an idea / need into a machine element or system which performs the required function safely, reliably and economically.

Machine member = any part (shaft, bolt, gear, spring, bearing ...) whose size and shape are decided by design.

Types of design

- Adaptive design - small changes in an existing design (most common).
- Development design - new design developed from an existing one, existing principle retained.
- New design - entirely new, needs original research (rarest).
- Also: rational design (stress analysis based), empirical design (formulae from experience), industrial design (looks, ergonomics, aesthetics), optimum design (best of all alternatives).

Design process (steps)

Recognition of need → Definition of problem → Synthesis
→ Analysis & optimisation → Evaluation (prototype / test)
→ Presentation (drawings, process sheets, reports)

- *Recognition of need*: a need / defect is felt (low efficiency, cost, noise, safety).
- *Definition*: specify input, output, speed, load, life, space, cost, environment.
- *Synthesis*: think of several possible mechanisms / schemes; select the best few.
- *Analysis*: forces, stresses, deflections -> size; iterate (trial & error) till requirement met.
- *Evaluation*: prototype testing, compare with specification; modify if needed.
- *Presentation*: working drawings with tolerances, material, heat treatment, assembly drawing, BOM.

2. Design Considerations

Factors a designer has to keep in mind while fixing the size and material of a member:

Group	Considerations
Strength / safety	type of load, stress, FOS, stiffness, deflection, vibration
Manufacture	casting / forging / machining, tolerance, surface finish, standard parts
Economy	material cost, labour, tooling, quantity, maintenance cost
Service	wear, corrosion, temperature, lubrication, life, reliability
Human / other	ergonomics, noise, safety guards, aesthetics, weight, space

Factor of safety (n or FOS) = failure stress / allowable (working) stress. For ductile material failure stress = S_y; for brittle = S_{ut}.

$$n = S_y / \sigma_{allow} \quad \text{(ductile)}$$
$$n = S_{ut} / \sigma_{allow} \quad \text{(brittle)}$$

(basic definition)

- Typical n: dead load steady 1.5-2 (ductile) ; live load 2-3 ; shock load 3-5 (up to 8-10 brittle).
- Too high n -> heavy & costly ; too low n -> risk of failure. Depends on: accuracy of load estimate, material scatter, consequence of failure, workmanship.

Note: Stiffness-governed members (machine tool bed, long shafts) are designed for deflection, not for strength alone. Check both.

3. Codes, Standards & Designation of Materials

Standard = set of specifications for size, material, quality, test, terminology, so that parts are interchangeable and safe. Code = set of rules for design / fabrication of a particular item (boiler code, pressure vessel code, AGMA for gears).

- Standards bodies: BIS (IS), ISO, ASME / ANSI, DIN, BS, AGMA, SAE, ASTM, JIS.
- Uses: interchangeability, reduction in variety of sizes, lower cost, better quality, safety, easy replacement, international trade.
- Preferred numbers (R5, R10, R20, R40 series - Renard) are used for standard sizes / speeds: ratio = 5-th root of 10 etc. (R10 -> 1.26 , R20 -> 1.12).

Designation of materials (Indian Standard, IS 1762 / IS 1570)

Material	Designation rule	Example & meaning
Grey cast iron	FG + min. tensile (N/mm^2)	FG 200 : S_{ut} = 200 MPa
Steel (by strength)	Fe + min. UTS	Fe 410 : S_{ut} = 410 MPa
Steel (by yield)	Fe E + min. yield	Fe E 250 : S_y = 250 MPa
Plain carbon steel	digits = C% × 100, C, Mn% × 100	45C8 : 0.45% C , 0.8% Mn
Alloy steel	C%×100, symbols, %×factor	40Ni14 : 0.4% C , 3.5% Ni
Alloy steel (2)	factor 4 : Cr,Ni,Mn,Si,W,Co	35Mn2Mo28 : .35%C, .5%Mn, .28%Mo
Malleable C.I.	BM / WM + UTS	BM 300 : black heart, 300 MPa
Cast steel	CS + UTS	CS 540 : S_{ut} = 540 MPa
Non-ferrous	chemical symbol + %	Cu Zn 40 : brass (60% Cu , 40% Zn)

Rule: Alloy steel: multiplying factor = 4 for Cr, Co, Ni, Mn, Si, W ; 10 for Al, Be, V, Pb, Cu, Nb, Ti, Ta, Zr ; 100 for P, S, N, C (in small amounts).

4. Selection of Materials

Material selection is balanced between properties required, processing and cost. Main criteria:

- Mechanical: strength (S_y, S_{ut}), ductility, hardness, toughness (impact), fatigue strength, creep strength, stiffness E.
- Physical: density (weight), thermal conductivity & expansion, melting point, electrical property.
- Manufacturing: castability, forgeability, machinability, weldability, heat-treatability.
- Service: corrosion / wear resistance, temperature, lubrication, noise damping (grey C.I. best damping).
- Economic: cost per kg, availability, standard sections, scrap value.

Member	Usual material	Reason
Shafts, axles	40C8, 45C8, 40Cr1	strength + toughness
Gears (heavy)	case-hardened 20MnCr5 , 40Ni14	hard teeth, tough core
Machine bed / housing	FG 200 , FG 260	castable, damping, cheap
Springs	55Si2Mn90 , music wire	high elastic limit
Bearing bushes	Bronze, babbitt	low friction, embeddability
Light parts	Al alloys	high strength / weight

5. Design for Static Loads

Static load = load that does not change with time (or changes very slowly). Member is designed so that the induced stress never exceeds the allowable stress.

Modes of failure

- Yielding (ductile) - permanent deformation makes the member unfit : S_y governs.
- Fracture (brittle) - sudden separation : S_{ut} governs ; also stress-raiser cracks.
- Excessive deflection - bad alignment, loss of accuracy (stiffness failure).
- Buckling - slender column / thin shell under compression (Euler).
- Fatigue - under repeated loads at stress < S_y ; creep - slow strain at high temperature ; wear, corrosion, fretting, impact, thermal shock.

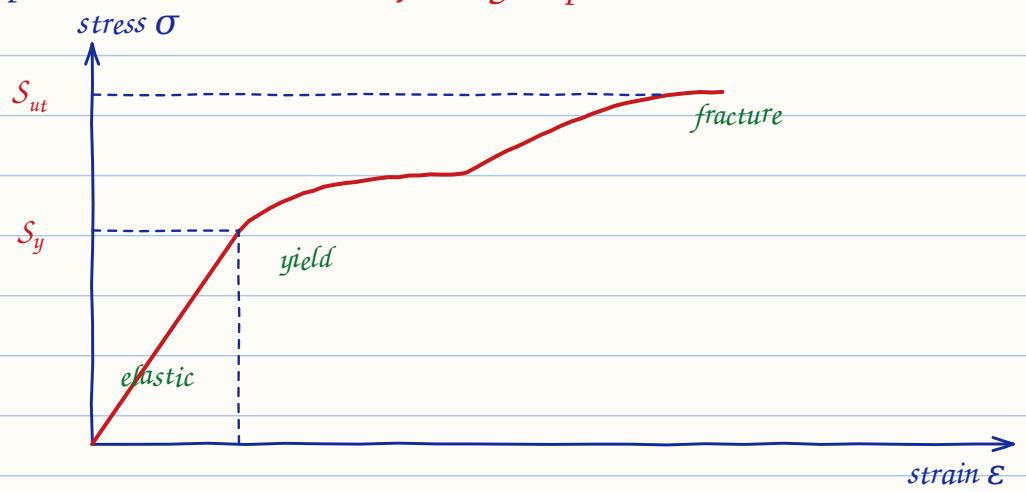


Fig 1: Stress-strain diagram of mild steel (ductile)

Ductile material (elongation > 5%) fails by yielding - design on S_y. Brittle material (C.I., ceramics) fails with no yield - design on S_{ut} (compression strength S_{uc} is much higher: $S_{uc} \approx 3\text{-}4\ S_{ut}$).

(a) Axial loading (tension / compression)

$$\sigma = P / A \le S_y / n$$

$$\text{solid round bar : } d = \sqrt{(4P / \pi \sigma_{allow})}$$

For a long column under compression also check buckling: $P_{cr} = \pi^2 E I / (K L)^2$. Direct shear (pins, keys, rivets): $\tau = P / A_{shear}$, $\tau_{allow} \approx S_{sy}/n$; $S_{sy} = 0.577\, S_y$ (distortion energy) or $0.5\, S_y$ (max shear).

(b) Bending

$$\sigma_b = M y / I = M / Z$$

$$\text{round : } \sigma_b = 32\, M / \pi d^3$$

$$\text{rectangle } (b \times h) : \sigma_b = 6\, M / b h^2$$

Transverse shear stress in beam (max): $\tau_{max} = 1.5\, V/A$ (rectangle), $4V/3A$ (solid circle), at neutral axis. Normally small compared with bending stress - check only for short, deep beams.

(c) Torsion

$$\tau = T r / J$$

$$\text{solid shaft : } \tau = 16\, T / \pi d^3$$

$$\text{hollow } (k = d_i/d_o) : \tau = 16\, T / \pi d_o^3 (1 - k^4)$$

Angle of twist $\theta = T L / (G J)$ rad ; torsional rigidity $G J$. Power : T (N·m) $= 60\, P / (2\pi N) = 9550\, P(kW) / N(rpm)$. Allowable $\tau = S_{sy}/n$; for shafts with keyway reduce by ~25%.

(d) Combined bending + torsion (shafts)

Principal stresses at the surface point where $\sigma = 32M/\pi d^3$ and $\tau = 16T/\pi d^3$ act:

$$\sigma_{1,2} = \sigma/2 \pm \sqrt{((\sigma/2)^2 + \tau^2)}$$

$$\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)} = 16 / \pi\, d^3 \cdot \sqrt{(M^2 + T^2)}$$

Define the equivalent twisting moment $T_e = \sqrt{(M^2 + T^2)}$ (Guest - ductile) and equivalent bending moment $M_e = \tfrac{1}{2} [M + \sqrt{(M^2 + T^2)}]$ (Rankine - brittle).

$$d^3 = 16\,T_e / \pi\,\tau_{allow}$$
$$d^3 = 32\,M_e / \pi\,\sigma_{allow}$$

(shaft diameter)

Example 1: Shaft carries M = 1000 N·m, T = 600 N·m. S_y = 300 MPa, n = 2. Find d.

$\tau_{allow} = S_y/2n = 300/4 = 75$ MPa (max shear theory). $T_e = \sqrt{(1000^2 + 600^2)} = 1166$ N·m.

$d^3 = 16 \times 1166\times10^3 / (\pi \times 75) = 79.2\times10^3$ mm³ → $d = 42.9$ mm → *provide d = 45 mm (standard).*

6. Impact (Shock) Loads

A load applied suddenly or falling from a height produces much larger stress than the same load applied gradually. Energy balance : work done by falling load = strain energy stored (½ $P_i\,\delta$).

Case	Equivalent static factor	Stress
Gradual load P	1	$\sigma_{st} = P/A$
Suddenly applied (h = 0)	2	$\sigma = 2\,P/A$
Falling weight W through h	$1 + \sqrt{(1 + 2h/\delta_{st})}$	$\sigma = \sigma_{st}(1 + \sqrt{(1 + 2h/\delta_{st})})$
Moving mass, velocity v	-	$\sigma = \sqrt{(\,m\,v^2\,E\,/\,A\,L\,)}$

$\delta_{st} = W\,L\,/\,A\,E$ (static deflection under W). For large h : $\sigma \approx \sigma_{st}\sqrt{(2h/\delta_{st})} = \sqrt{(2\,W\,h\,E\,/\,A\,L)}$.

- Impact stress can be reduced by increasing L (long bolt), reducing A in unthreaded parts (reduced shank), using springs / washers - all increase the strain energy absorbed.

Example 2: W = 1 kN falls h = 50 mm on a collar of a bar L = 1 m, d = 20 mm, E = 200 GPa. A = 314 mm², $\delta_{st} = 1000\times1000/(314\times200000) = 0.0159$ mm, $\sigma_{st} = 3.18$ MPa.

$\sigma = 3.18\,[\,1 + \sqrt{(1 + 2\times50/0.0159)}\,] = 3.18\,(1 + 79.3) \approx 255$ MPa (80 times static !).

7. Theories of Failure (static loads)

A theory predicts failure under combined (biaxial / triaxial) stress from the simple tension test value. Principal stresses $\sigma_1 \geq \sigma_2 \geq \sigma_3$; for plane stress:

$$\sigma_{1,2} = (\sigma_x+\sigma_y)/2 \pm \sqrt{\left((\sigma_x-\sigma_y)/2\right)^2 + \tau_{xy}^2}$$

Theory (name)	Failure criterion	Design equation	Best for
Max principal stress (Rankine)	$\sigma_1 = S_y$ or S_{ut}	$\sigma_1 \le S/n$	brittle
Max shear stress (Tresca / Guest)	$\tau_{max} = S_y/2$	$\sigma_1 - \sigma_3 \le S_y/n$	ductile (conservative)
Max principal strain (St. Venant)	$\varepsilon_1 = S_y/E$	$\sigma_1 - \mu\sigma_2 - \mu\sigma_3 \le S_y/n$	(rarely used)
Max strain energy (Haigh)	$U = S_y^2/2E$	$\sigma_1^2+\sigma_2^2-2\mu\sigma_1\sigma_2 \le (S_y/n)^2$	(rarely used)
Max distortion energy (von Mises-Hencky)	$\sigma' = S_y$	$\sigma' = \sqrt{(\sigma_1^2-\sigma_1\sigma_2+\sigma_2^2)} \le S_y/n$	ductile (best fit)

$$\text{von Mises : } \sigma' = \sqrt{\sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2}$$
$$\text{bending + torsion : } \sigma' = \sqrt{\sigma^2 + 3\tau^2}$$

(DET)

- Pure shear, ductile : Tresca $S_{sy} = 0.5\ S_y$; von Mises $S_{sy} = 0.577\ S_y$ (experiments agree with 0.577).
- Tresca is safer (hexagon inside the von Mises ellipse) ; difference at most 15%.
- Brittle material : use Rankine (or Coulomb-Mohr when $S_{uc} \neq S_{ut}$) : $\sigma_1/S_{ut} - \sigma_3/S_{uc} = 1/n$.

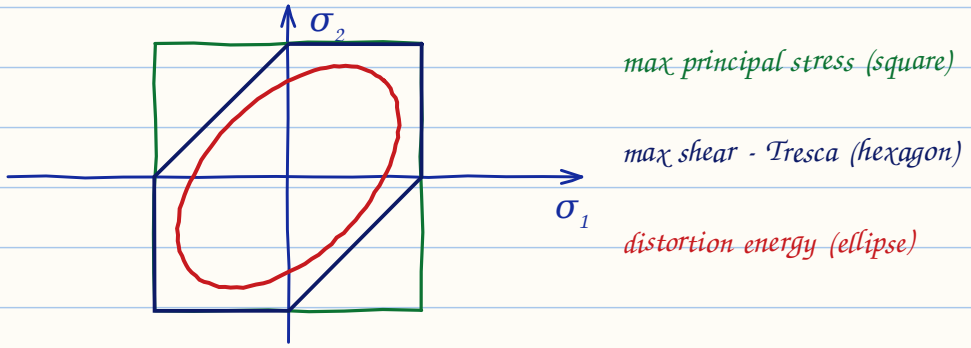


Fig 2: Failure loci in the σ_1 - σ_2 plane

8. Design for Dynamic (Fatigue) Loads

Fatigue : failure of a member under repeated / fluctuating stress at a value well below S_{ut} (even below S_y), after a number of cycles N. About 90% of service failures. Starts as micro-crack at a stress raiser → propagates (beach marks) → sudden final fracture of the remaining area.

Endurance limit & S-N curve (R. R. Moore test)

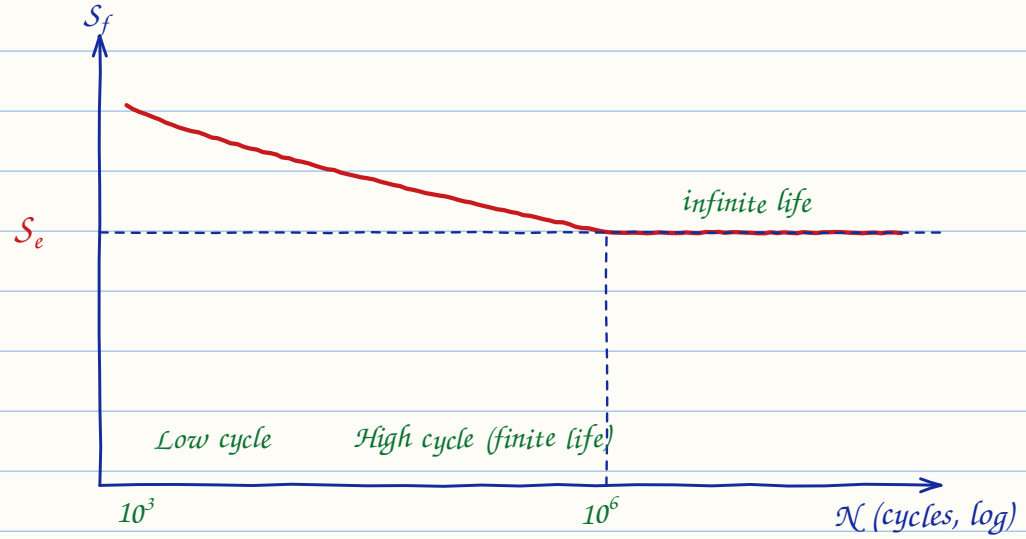


Fig 3: S-N curve for steel (knee at 10^6 cycles)

- Endurance limit S_e' = max completely reversed stress for infinite life ($N > 10^6$) of a standard polished specimen (7.5 mm dia).
- Steels show a knee ; non-ferrous (Al, Cu) have no knee - fatigue strength quoted at 10^8-5×10^8 cycles.

$$S_e' = 0.5\, S_{ut} \quad (S_{ut} \le 1400\ MPa)$$
$$S_e' = 700\ MPa \quad (S_{ut} > 1400\ MPa)$$

(steel, bending)

Also : cast iron $S_e' \approx 0.4\, S_{ut}$; Al alloys $S_e' \approx 0.4\, S_{ut}$ (≤ 130 MPa). Fatigue strength at 10^3 cycles $\approx 0.9\, S_{ut}$ (bending), $0.75\, S_{ut}$ (axial). Between 10^3 and 10^6: log S_f varies linearly with log N (Basquin), $S_f = a\, N^b$.

Modified (corrected) endurance limit - Marin factors

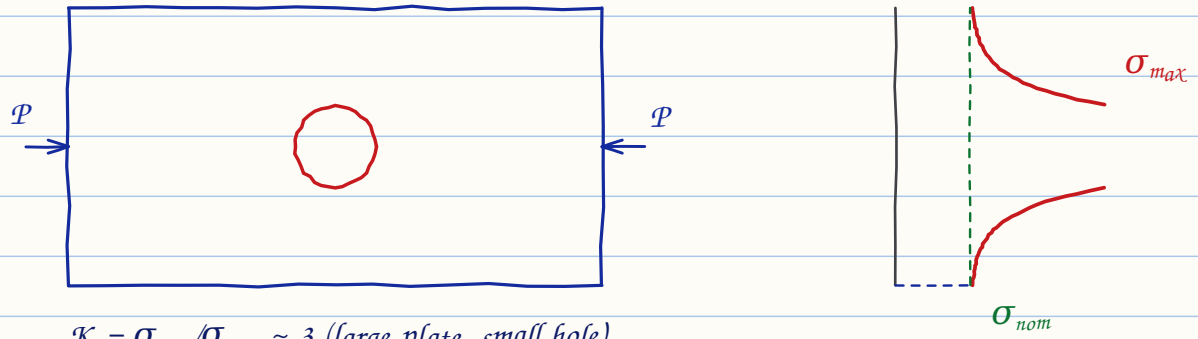
$$S_e = k_a\, k_b\, k_c\, k_d\, k_e \cdot S_e'$$

Factor	Meaning	Typical value
k_a	surface finish (mirror 1.0 ; ground 0.9 ; machined 0.78 ; hot rolled 0.6 ; forged 0.5)	0.5 - 1
k_b	size : 1.24 $d^{-0.107}$ (2.79-51 mm) ; axial load → 1 ; Bhandari : 1 (d<7.5), 0.85 (7.5-50), 0.75 (>50)	0.7 - 1
k_c	load type : bending 1.0 ; axial 0.85 ; torsion 0.59	0.59 - 1
k_d	temperature : 1 at room T, falls above 450°C	≤ 1
k_e	reliability : 50% 1.0 ; 90% 0.897 ; 95% 0.868 ; 99% 0.814 ; 99.9% 0.753	0.75 - 1

For torsion (ductile) use $S_{se} = 0.577\ S_e$ (or simply apply $k_c = 0.59$ to S_e'). Shear yield $S_{sy} = 0.577\ S_y$.

9. Stress Concentration & Notch Sensitivity

Holes, shoulders, keyways, grooves, threads, sharp corners → abrupt change of section → local stress much higher than nominal (flow of force lines crowd together).



$K_t = \sigma_{max}/\sigma_{nom} \approx 3$ (large plate, small hole)

Fig 4: Stress concentration at a hole

$$K_t = \sigma_{max} / \sigma_{nom} \quad \text{(theoretical, from geometry only)}$$
$$K_f = 1 + q\,(K_t - 1) \quad \text{(fatigue stress conc. factor)}$$
$$q = (K_f - 1)/(K_t - 1) \quad , \quad 0 \le q \le 1$$

- Notch sensitivity q : $q = 0$ → material insensitive (K_f=1) ; $q = 1$ → fully sensitive (K_f=K_t). Depends on material and notch radius r : larger r , higher S_{ut} → larger q. Cast iron $q \approx 0.2$ (internal flaws) ; hardened steel $q \to 1$.
- Static loads : ductile material - K_t is ignored (local yielding relieves it) ; brittle material - K_t must be applied. Fatigue : K_f is always applied (on alternating stress).

Methods of reducing stress concentration

- Generous fillet radius at shoulders ; relief grooves ; undercuts ; gradual taper.
- Additional (stress-relief) notches / holes on the shoulder that make force flow smooth ; avoid sharp keyways at critical sections - use profile (sled-runner) keyways.
- Cold working, shot peening, surface rolling → compressive residual stress, improves fatigue strength.

10. Types of Fluctuating Loads

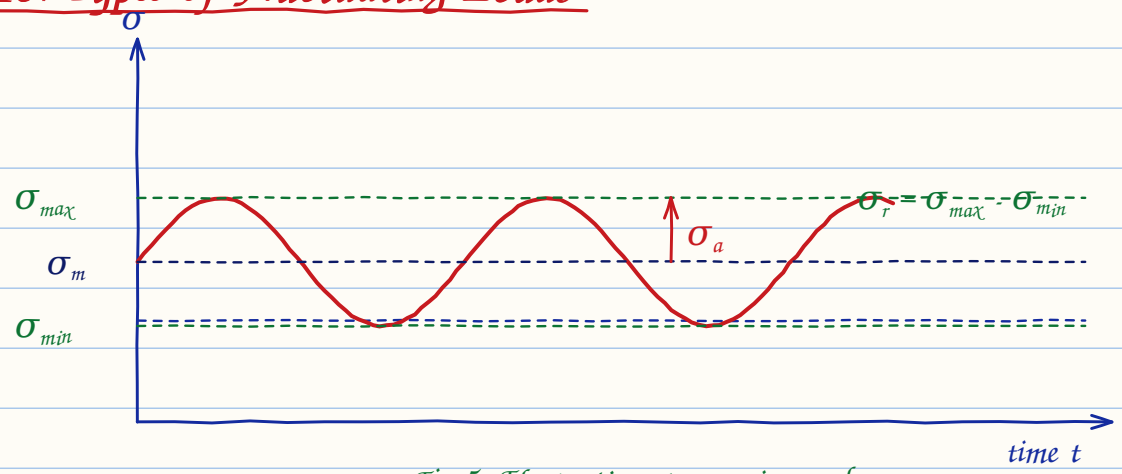


Fig 5: Fluctuating stress - time cycle

$$\sigma_m = (\sigma_{max} + \sigma_{min})/2 \quad , \quad \sigma_a = (\sigma_{max} - \sigma_{min})/2$$

$$\mathcal{R} = \sigma_{min}/\sigma_{max} \quad , \quad \mathcal{A} = \sigma_a/\sigma_m$$

Type	σ_m	σ_a	$\mathcal{R}$
Completely reversed (rotating shaft)	0	σ_{max}	-1
Repeated / pulsating (gear tooth)	$\sigma_{max}/2$	$\sigma_{max}/2$	0
General fluctuating (spring, bolt)	$\neq 0$	$\neq 0$	$-1 < \mathcal{R} < 1$
Steady (static)	σ	0	+1

11. Fatigue Design for Infinite Life

For a member under σ_m and σ_a the (σ_m, σ_a) point must lie below the failure line of the Haigh / Goodman diagram. Endpoints : $\sigma_a = S_e$ on the vertical axis ($\sigma_m = 0$), $\sigma_m = S_{ut}$ or S_y on the horizontal axis.

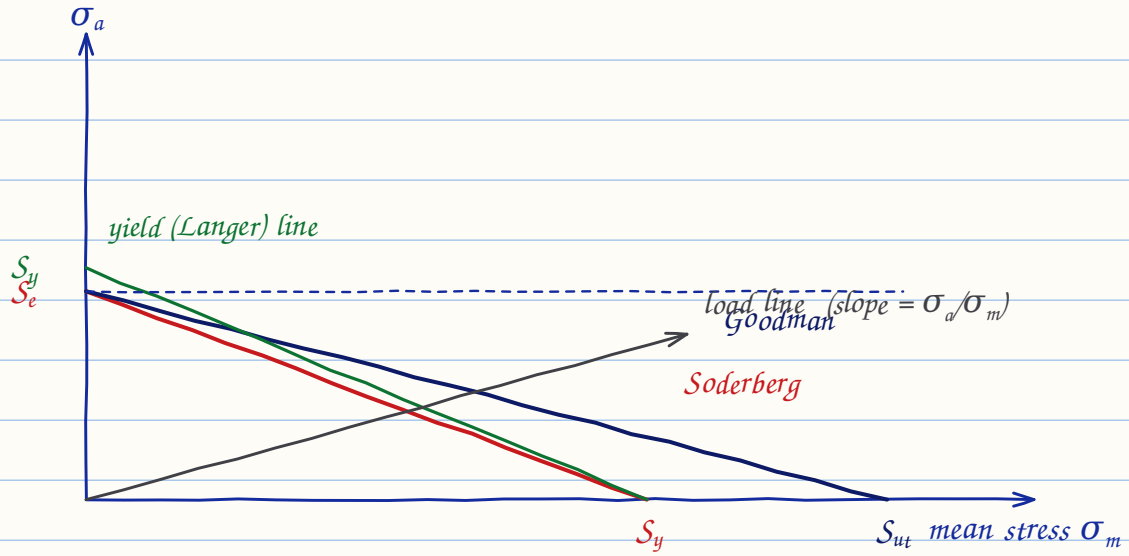


Fig 6: Failure criteria on the σ_m - σ_a diagram

Criteria (n = factor of safety)

Soderberg : $\sigma_a/S_e + \sigma_m/S_y = 1/n$

Goodman : $\sigma_a/S_e + \sigma_m/S_{ut} = 1/n$

Gerber : $n\,\sigma_a/S_e + (n\,\sigma_m/S_{ut})^2 = 1$

- Soderberg - most conservative (line to S_y ; also guards against yielding) ; used when safety is vital.
- Goodman - fits brittle-material data better ; straight line S_e to S_{ut}.
- Modified Goodman - Goodman line + a yield (Langer) line $\sigma_a + \sigma_m = S_y$ to check first-cycle yielding. Design point must lie inside both.
- Gerber parabola - best fit to ductile test data (average, less safe) ; Gerber / Goodman are used for reliable loads.
- Stress concentration : use K_f on the alternating part ($\sigma_a = K_f\,\sigma_{a,nom}$) ; mean stress normally not multiplied for ductile steels ($K_{fm} = 1$) but is for brittle ones.

Remember: Use S_e (corrected, with k_a...k_e) and K_f in every fatigue equation. In axial / torsion cases replace S_e, S_y, S_{ut} by S_{se}, S_{sy}, $S_{su} = 0.577\,S_y$, $0.8\,S_{ut}$.

Example 3: A steel bar (S_{ut} = 600, S_y = 380 MPa) has S_e = 220 MPa (all corrections) and K_f = 1.5. Axial load varies 10 kN to 30 kN. Find d for n = 2 by Soderberg.

P_m = 20 kN , P_a = 10 kN. $\sigma_m = 20000/A$, $\sigma_a = 1.5 \times 10000/A$.

$1/2 = (15000/A)/220 + (20000/A)/380 = (68.2 + 52.6)/A \rightarrow A = 241.6\ mm^2 \rightarrow d = \sqrt{(4A/\pi)} = 17.5 \rightarrow d = 18\ mm.$

(Goodman with the same data : $A = 203\ mm^2 \rightarrow d = 16.1\ mm$, i.e. less conservative.)

12. Fatigue Design under Combined Stresses

Shafts carry bending (σ) and torsion (τ) that are each fluctuating : σ_m, σ_a, τ_m, τ_a. Use the distortion-energy theory to convert them to an equivalent von Mises mean and alternating stress, then apply Soderberg / Goodman.

$$\sigma_a' = \sqrt{\left((K_f \sigma_a)^2 + 3\,(K_{fs} \tau_a)^2 \right)}$$
$$\sigma_m' = \sqrt{\left(\sigma_m^2 + 3\,\tau_m^2 \right)}$$

$$1/n = \sigma_a'/S_e + \sigma_m'/S_y \quad \text{(Soderberg)}$$
$$1/n = \sigma_a'/S_e + \sigma_m'/S_{ut} \quad \text{(Goodman)}$$

- S_e here is the (bending) endurance limit with k_a, k_b, k_d, k_e ; the torsion effect $k_e = 0.59$ is already taken by the $3\,\tau_a^2$ term - do not apply it twice.
- Rotating shaft, steady torque T and bending M : $\sigma_a = 32\,M/\pi d^3$, $\sigma_m = 0$, $\tau_m = 16\,T/\pi d^3$, $\tau_a = 0 \rightarrow \sigma_a' = K_f\,32M/\pi d^3$, $\sigma_m' = \sqrt{3} \cdot 16T/\pi d^3$.

$$(32\,n / \pi\,d^3) \cdot \sqrt{\left((K_f\,M / S_e)^2 + (3/4)(T / S_y)^2 \right)} = 1 \quad \text{(Soderberg - shaft)}$$

- Steps : (1) loads & reactions → M, T diagrams ; (2) critical section ; (3) choose material → S_{ut}, S_y ; (4) S_e', Marin factors → S_e ; (5) K_t, q → K_f ; (6) assume d, find σ_a, σ_m ; (7) apply criterion → d ; (8) round to standard size & check.

Finite life (quick note)

If life N is between 10^3 and 10^6 : $S_f = a\,N^b$, $a = (f\,S_{ut})^2/S_e$, $b = -(1/3)\log_{10}(f\,S_{ut}/S_e)$, $f \approx 0.9$; use S_f instead of S_e in the criterion. Cumulative damage (Miner) : $\Sigma\ n_i/N_i = 1$.

Quick revision - Unit I

Topic	Key formula / idea
Design stress	$\sigma_{allow} = S_y/n$ (ductile) ; S_{ut}/n (brittle)
Shaft (static)	$d^3 = 16\sqrt{(M^2+T^2)} / (\pi\ \tau_{allow})$
Impact	$\sigma = \sigma_{st}[1 + \sqrt{(1 + 2h/\delta_{st})}]$
von Mises	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)} \leq S_y/n$
Endurance limit	$S_e = k_a k_b k_c k_d k_e (0.5\ S_{ut})$
Notch	$K_f = 1 + q(K_t - 1)$
Fatigue (infinite life)	$\sigma_a/S_e + \sigma_m/S_{y\ or\ ut} = 1/n$